Problem Summary

- Given the root of a binary tree, determine whether the tree is symmetric around its center. A tree is symmetric if the left subtree is a mirror of the right subtree.

Key Idea

To check symmetry, compare two nodes at the same time.

For two trees to be mirrors:

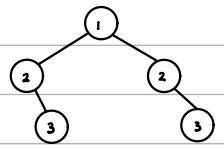
→ their values must be equal

→ left of first must match right of second

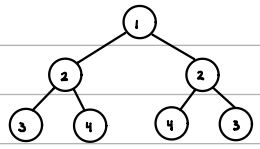
→ right of first must match right of second

Diagram Examples

Non Symmetric:



Symmetric:



Approach

→ DFS Recursion

1) If root is empty, return True

2) Define helper function isMirror(t_1, t_2)

3) In helper

→ if both nodes are None, return True

→ if one is missing or values differ, return False

→ recursively compare mirrored children

4) Start by comparing: root.left and root.right

Time Complexity: $O(n)$ / $O(\log n)$ → Each node is compared once
(Skewed Tree / Balanced Tree)

Space Complexity: $O(h)$ → Recursion stack, where h is tree height